

Toward integration of Behavioral and Semantic Interoperability Testing in Energy Data Spaces

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Abstract

Data spaces provide a federated environment for trusted data exchange among heterogeneous and autonomous actors while preserving data sovereignty and governance rules. In such environments, syntactic and semantic interoperability are ensured in emerging implementations by using harmonized metadata and shared ontologies and by performing constraint-based validation of exchanged information. However, this level of conformance alone does not ensure that participating systems execute required actions in a behaviorally conformant way. In an Energy Data Space, this limitation may become an issue for coordinated services such as local energy production and consumption management, household appliances flexibility optimization and congestion management. This paper proposes a conceptual ontology-driven framework for behavioral interoperability testing that complements syntactic and semantic conformance checking with validation of dynamic behavior. The framework builds on RDF and SHACL-based models and outlines how explicit validation of state transitions and temporal constraints can be incorporated alongside semantic conformance checking. Its applicability within an Energy Data Space is illustrated in simplified form through an energy flexibility management use case following the European Joint Research Centre (JRC) Code of Conduct (CoC) for Energy Smart Appliances (ESA).

Keywords

Behavioral Interoperability, Semantic Interoperability, Ontology-Driven Constraint Testing, Data Space, Energy Data Space, Energy Flexibility Management, RDF, SHACL,

This paper contributes to Challenge 2.4 (Interoperable Policy Engines) of the W3C Dataspaces Community Group by introducing ontology-driven behavioral interoperability testing.

1. Introduction

Energy Data Space initiatives aim to enable interoperable data exchange among independent actors in the energy sector while preserving data sovereignty and governance requirements [1]. More broadly, data space initiatives are progressing from architectural blueprints (e.g., Data Spaces Support Centre (DSSC) [2]) toward implementation in sectoral domains such as the European Mobility Data Space [3] and the ongoing development of the Common European Energy Data Space (CEEDS) [4] under the European Data Strategy. Within this landscape, semantic assets, such as shared ontologies, vocabularies, and associated constraints, provide a basis for common representation and interpretation of exchanged data [5]. In the context of the European Joint Research Centre (JRC) Code of Conduct (CoC) for Energy Smart Appliances (ESA) [6] and of the development of Energy Data Spaces, the provision and consumption of interoperable data are essential to enable coordination among energy smart appliance manufacturers, energy management systems, prosumers, grid operators, energy suppliers, and aggregators. By exposing energy smart appliances capabilities, real-time status information, flexibility offers and grid conditions

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using harmonized metadata (e.g., based on DCAT-AP [7] and EN 304 199 [8]) and semantically enriched data (e.g., based on SAREF [9] and EN 303 760 [10]), different actors can discover, interpret, and integrate each other's datasets. In such environments, shared semantic models that formalize domain concepts and their relationships play a critical role in enabling interoperability [11]. This corresponds to the semantic facet (e.g., layer) of interoperability, which is one of the five facets (Transport, Syntactic, Semantic, Behavioral, and Policy) described in ISO/IEC 21823-1:2019 [12]. In practice, implementations often validate syntactic conformance together with semantic conformance, since semantic validation relies on structurally well-formed payloads that can be parsed and interpreted against a shared model. Focusing on the semantic facet, ISO/IEC 21823-3:2021 defines semantic interoperability as *"interoperability so that the meaning of the data model within the context of a subject area is understood by the participating systems"* [13]. Emerging data space implementations [14] represent semantic models using the Resource Description Framework (RDF) [15] and apply constraint languages such as Shapes Constraint Language (SHACL) [16] to assess syntactic and semantic conformance (i.e., exchanged RDF graph satisfies the ontology-defined semantic constraints used for validation, e.g., SHACL shapes) of exchanged data. Building on this semantic foundation, semantic interoperability testing verifies whether two or more systems interpret and reason over exchanged data consistently within a shared semantic context [17, 18]. This paper recognizes the essential role of semantic interoperability testing in data space implementations by (i) building on an Ontology-Driven Constraint (ODC) Tester that performs semantic conformance checking through SHACL-based validation of RDF payloads, and (ii) extending it with a complementary behavioral interoperability perspective.

Within the Int:net¹ eu project, we developed a proof-of-concept of the ODC-Tester², which performs SHACL-based constraint validation of RDF data against reference ontologies such as SAREF [9] and SAREF4ENER [19] to enable semantic conformance checking [20]. However, syntactic and semantic conformance do not by themselves ensure that interacting systems produce the expected operational behavior during execution. Here, operational behavior refers to the observable execution of control actions during interaction, including the applied control rules, execution conditions, and timing choices that determine the resulting outcomes. ISO/IEC 21823-5 defines behavioral interoperability as *"interoperability so that the actual result achieves the expected outcome"* [21]. Behavioral interoperability testing therefore validates whether interacting systems or components exhibit coordinated and expected behaviors in a shared operational environment, by validating conformant execution through synchronization, state conformance, adherence to defined behavioral rules, and achievement of intended operational outcomes [22, 23]. Behavioral interoperability is particularly relevant for coordinated Energy Data Space services (e.g., local energy production and consumption management, household appliance flexibility optimization, and congestion management), where semantically conformant data exchanges are expected to trigger correct and timely control actions under shared governance and operational constraints. Therefore, interoperability assurance also requires verifying operational interaction rules, including adherence to the required order of actions and to specific timing constraints. This verification is currently performed through functional testing, which is often time-consuming and may lead to errors or oversights, especially for complex workflows involving multiple interacting systems in Energy Data Space contexts. Consequently, this paper addresses the following research question:

RQ: How can behavioral interoperability be systematically tested in a semantic interoperability environment that represents data through ontologies?

To address this question, ontology-driven semantic conformance checking is complemented with behavioral interoperability testing capabilities. The result is a conceptual framework that supports the validation of expected operational behavior alongside semantic conformance checking. The main contribution of this work is the conceptual integration of semantic and behavioral interoperability testing, with the aim of supporting interoperability assurance in Energy Data Space implementations.

¹Int:net eu project: <https://intnet.eu>

²Zenodo repository of the ODC-Tester: <https://zenodo.org/records/13955566>

The paper is structured as follows: in Section 2, we discuss the limits of semantic conformance and clarify the need for behavioral interoperability; in Section 3, we review related work; in Section 4, we propose our framework for ontology-driven behavioral interoperability testing; in Section 5, we conclude the paper and present directions for future work.

2. Background and Motivation

Building on the limitation identified in Section 1, we enhance semantic conformance with behavioral conformance. Semantic conformance concerns whether exchanged information is interpreted according to the agreed semantic definitions. Behavioral conformance concerns whether the interaction, when executed by independent participants, leads to the expected outcome under the agreed interaction rules. Figure 1 highlights that a comprehensive interoperability assessment therefore needs to address not only shared meaning, but also the operational result of the interaction.

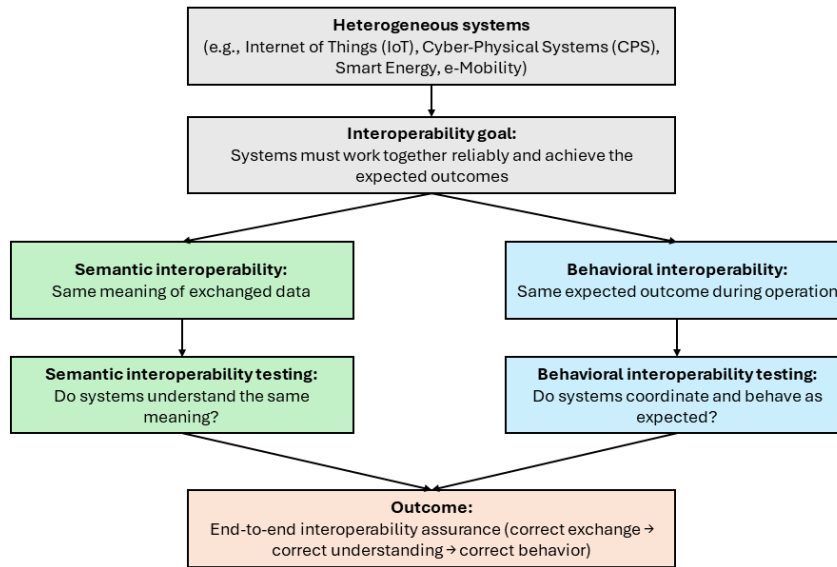


Figure 1: A Complementary Perspective Integrating Semantic and Behavioral Interoperability Assurance

From a testing perspective, bridging the gap between semantic and behavioral interoperability presents three main challenges. First, semantically conformant data exchanges can still produce non-conformant outcomes because interacting systems (e.g., smart appliances or EMS components) may apply different control logic, state assumptions, or timing choices during operation. Second, governed interactions rely on agreed rules (e.g., obligations, ordering, and time bounds) that constrain how these systems are expected to act across exchanges, and these are not ensured by semantic conformance alone. Third, behavioral conformance requires trace-based evidence of what happened during interaction, since outcome-oriented assurance depends on the observed ordering and timing of events. These challenges motivate behavioral interoperability testing as a complement to semantic interoperability testing.

3. Related Work

Behavioral aspects in ontology-based systems have been explored in different research contexts. Existing research demonstrates how dynamic behavior can be represented or validated using semantic technologies. However, these approaches are generally not explicitly framed within interoperability assurance workflows for distributed data space interactions. In data space environments, semantic conformance is often assessed over exchanged semantic payloads (e.g., using SHACL or similar constraint languages), while behavioral assurance requires trace evidence across message exchanges and time.

Larhrib et al. [24] model executable behavioral flows directly in RDF using SHACL and SPARQL. Tasks, states, and transitions are represented as ontology elements and validated during execution. Silva et al. [23] use an ontology to structure behavior-driven testing scenarios for interactive systems. Both works demonstrate that ontologies can support validation beyond static schema checking. Their main goal is to validate workflows or application behavior. They do not focus on interoperability validation between autonomous systems exchanging semantic data.

Chen et al. [25] translate Systems Modeling Language (SysML) behavioral models into Web Ontology Language (OWL) and apply ontology-based reasoning to validate behavioral conformance at design time. This approach strengthens model correctness before deployment. The validation is performed on design artifacts and does not evaluate runtime interaction traces between operational systems.

Temporal and concurrency constraints have also been addressed through hybrid semantic formal approaches. Liu et al. [26] introduce a Semantic Petri Net model that combines Colored Petri Nets with RDF to validate ordering and timing constraints in digital twin environments. Al Shami et al. [27] propose ReqTLo (Requirements Temporal Logic Ontology), which embeds Metric Temporal Logic within an ontology to support time-bounded requirement checking in simulation-based environments. These approaches provide explicit temporal semantics and formal analysis capabilities. However, behavioral validation is commonly realized through an additional formalism or tooling layer (e.g., Petri Nets or temporal logic engines) rather than being explicitly structured as part of an ontology-grounded interoperability validation workflow over exchanged data space data.

The literature therefore shows substantial progress in semantic validation, behavioral modeling, and temporal rule checking. At the same time, these aspects are often developed as separate components. Their systematic integration into a unified, ontology-driven interoperability assurance process for distributed system interaction remains limited.

This paper contributes at this integration point by proposing a conceptual workflow for interoperability assurance based on exchanged traces that combines ontology-driven semantic conformance validation with trace-based behavioral validation. Rather than addressing ontology-based workflow validation, design-time behavioral reasoning, or temporal/formal validation in isolation, the framework focuses on evaluating exchanged traces against instantiated execution and timing constraints in Energy Data Space interactions. In this way, it provides a structured basis for incorporating behavioral conformance checking alongside semantic conformance validation in interoperability assurance processes.

4. Ontology-Driven Behavioral Interoperability Testing Framework

To address the limitation and the gap identified in Section 3, we extend the ODC-Tester developed in our previous work [20] with behavioral testing capabilities. In our extension, we propose an ontology-driven behavioral interoperability testing framework that supports trace-based evaluation of dynamic behavior.

Figure 2 illustrates the overall testing capability of the ODC-Tester. Conformity Aspects of Interest (CAIs) refer to the aspects selected for conformance assessment against a reference ontology and are processed by the Ontology-Driven Constraint Testing component (represented as a square box in figure 2), which orchestrates the validation workflow through two integrated testing components: semantic interoperability testing and behavioral interoperability testing. The validation outcomes of both layers are forwarded to the ODC Test Engine (Evidence Aggregation), which produces a unified validation report together with the execution trace, including recommendations. This integrated workflow is intended to support the combined assessment of semantic conformance and behavioral conformance within a single testing process. Following this integrated semantic and behavioral testing workflow, which constitutes the main conceptual contribution of this paper, we provide details of the proposed framework for behavioral interoperability testing within the ODC-Tester.

Figure 3 presents the proposed behavioral interoperability testing framework and its execution workflow. The behavioral layer is organized around three main components: the Behavioral Test Manager, the Time Manager, and the Flow Manager. Two runtime artefacts support execution and

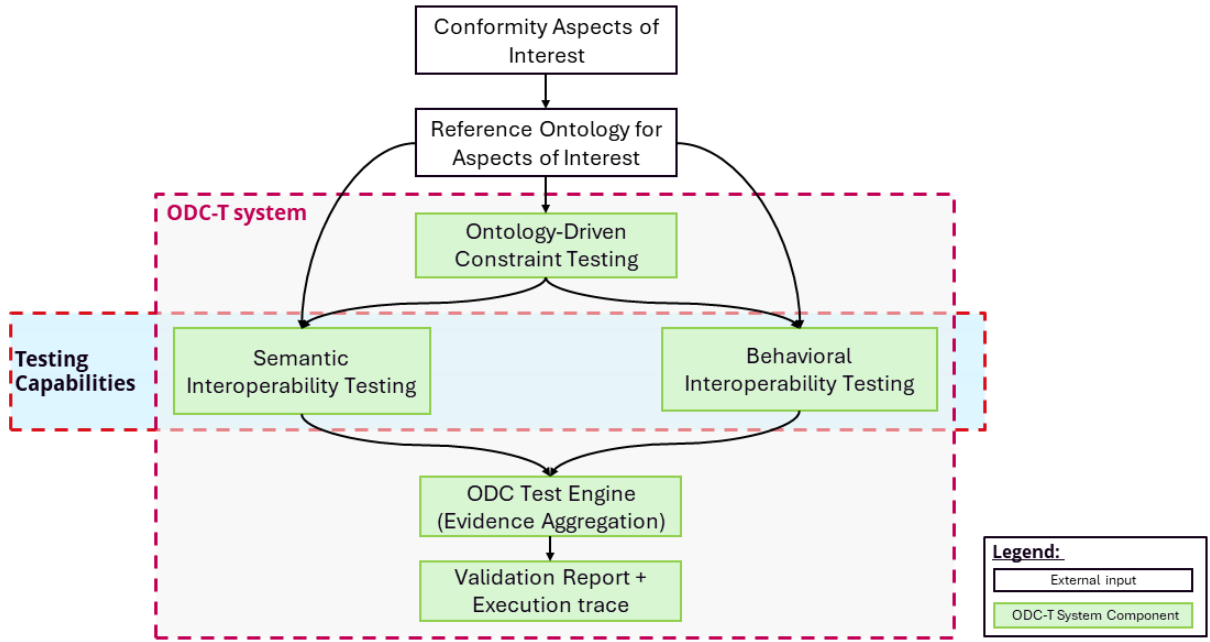


Figure 2: Overall testing capability of the ODC-Tester

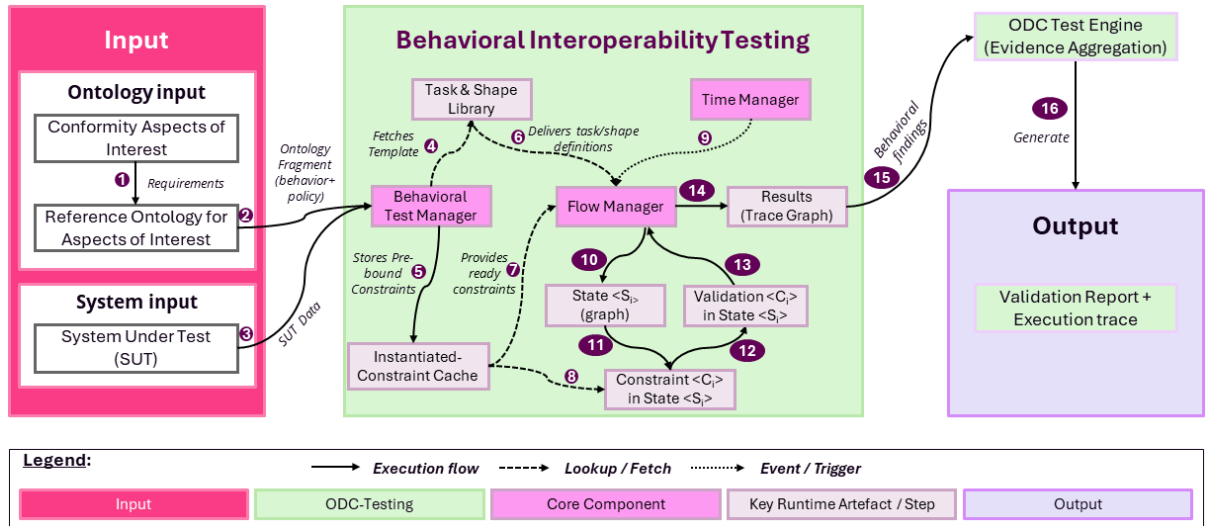


Figure 3: Proposed behavioral interoperability testing framework and workflow.

inspection: the Instantiated Constraint Cache and the RDF Trace Graph.

For clarity, the workflow is structured into three sequential phases, referenced by the step numbers in Figure 3: (i) input and instantiation (Steps 1-6), (ii) execution and validation (Steps 7-13), and (iii) evidence aggregation and reporting (Steps 14-16). Execution begins when a tester selects a CAI (Conformity Aspects of Interest) specifies the behavioral requirement under evaluation (Step 1). The selected CAI is then aligned with a reference ontology chosen for the evaluation context in order to derive an ontology fragment capturing the relevant behavioral and policy terms (Step 2). In parallel, the System Under Test (SUT) provides runtime data required to bind the evaluation context (Step 3). The Behavioral Test Manager retrieves the suitable behavioral template from the Task and Shape Library (Step 4). The template is represented as an RDF graph describing tasks, states, transitions, and associated constraint definitions. It is instantiated for the current session by binding placeholders using values from the

ontology fragment and SUT context, resulting in session-specific flow artefacts and SHACL/SPARQL constraint definitions stored in the Instantiated Constraint Cache (Step 5). The Task and Shape Library further provides task and shape definitions required during execution (Step 6). During execution, the Flow Manager retrieves and applies the instantiated constraints and flow elements from the cache (Steps 7-8). It maintains an evolving RDF state graph, denoted as S_i (Step 10), representing the current execution snapshot. For each state S_i , the applicable constraint set C_i is selected from the Instantiated Constraint Cache (Step 11) and validated in the current state context (Step 12). SHACL validation and SPARQL-based operations are applied to evaluate preconditions, execute transitions, and validate postconditions. The Time Manager provides explicit temporal semantics required for time-bounded evaluation (Step 9). It emits tick and deadline events that are integrated into execution to support checks involving time windows, maximum durations, and timeout conditions. Temporal events are treated as part of the RDF-based execution context rather than as external procedural logic. Validation outcomes determine flow progression and local transition verdicts (Step 13). All behavioral outcomes, including validation outcomes, state identifiers, applied constraints, events, and timestamps, are written to the RDF Trace Graph (Step 14). The Trace Graph is then forwarded to the ODC Test Engine (Evidence Aggregation) (Step 15), which aggregates semantic conformance outcomes and behavioral trace evidence to generate a unified validation report with trace-based evidence attached (Step 16). The workflow described above outlines the end-to-end behavioral interoperability testing process.

To illustrate the proposed framework, we use a simplified use case inspired by the JRC CoC for Energy Smart Appliances (ESA) [6]. We follow the Flexible Start use case Scenario 3 (Select alternative power sequence) using a household appliance (dishwasher). This example is presented in simplified form for demonstration purposes. Broader interaction scenarios are not addressed here, and appliance behavior, requirements, and timing constraints may vary across real implementations.

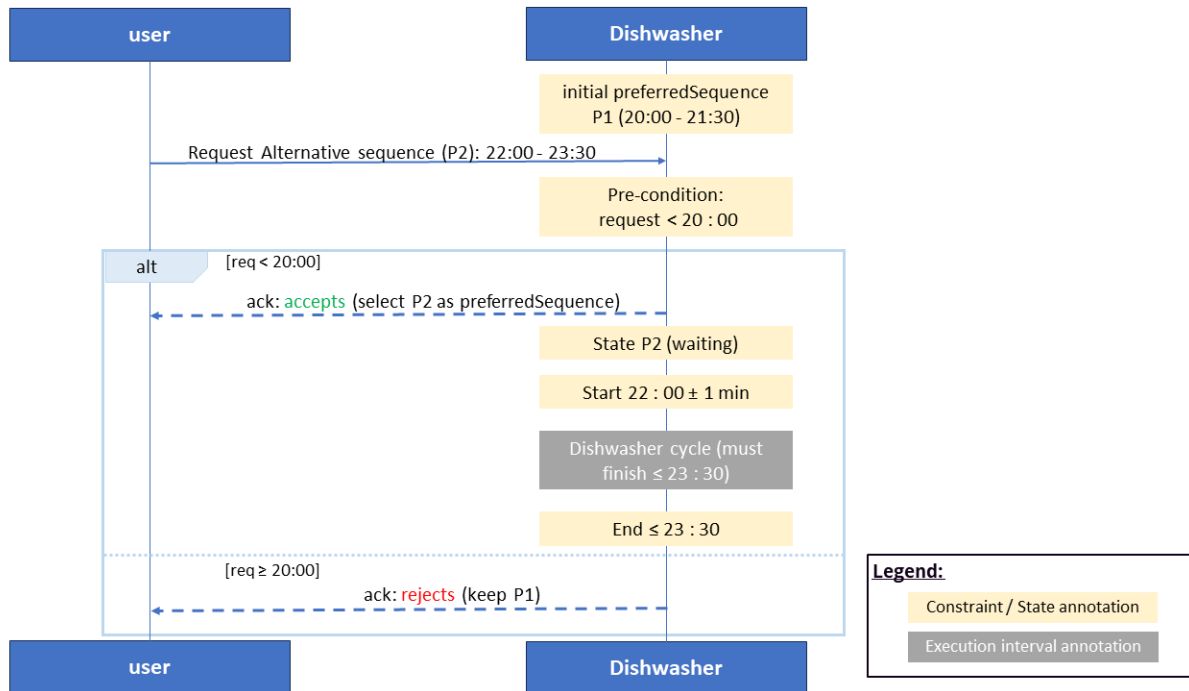


Figure 4: Sequence diagram for alternative power sequence selection (Dishwasher) derived from [6].

Figure 4 presents the sequence diagram for the selected example use case (select alternative power sequence). After the dishwasher announces a plan with an initial preferred sequence $P1$ (20:00-21:30) and an optional sequence $P2$ (22:00-23:30), a user (acting as a Customer Connectivity Manager (CCM) in this example) requests selecting $P2$ and configures the corresponding start time. The sequence diagram (figure 4) makes explicit the key decision point: the request is accepted only if it is received

before 20:00; otherwise it is rejected and $P1$ is kept. When accepted, the dishwasher is expected to (i) enter a waiting state for $P2$, (ii) start execution at $22:00 \pm 1$ minute, and (iii) complete the cycle by 23:30. The annotations in the diagram indicate the state/constraint checkpoints and the execution interval that must be verified from observable evidence (timestamps and state changes).

Table 1 derives a compact set of behavioral rules from Figure 4 and specifies the trace evidence required for behavioral interoperability testing. The rules are stated as behavioral test criteria and cover (i) the schedule constraint at request time (request received before 20:00), (ii) the post-acceptance state transition (enter waiting state for $P2$), (iii) the start-time constraint ($22:00 \pm 1$ minute), and (iv) the completion constraint (finish by 23:30). For each rule, the table indicates the observable trace elements used for validation (e.g., request timestamp, acceptance outcome, state transition, and observed start/end timestamps). These rules provide the behavioral test basis used by the proposed framework to produce a traceable behavioral verdict.

Table 1

Behavioral rules and required trace evidence for the alternative power sequence selection use case (Dishwasher).

Behavioral rule	Evidence to check and framework component for validation
Accept request to select $P2$ only if the request is received before 20:00.	<i>Observable evidence:</i> Request timestamp < 20:00. <i>Framework component:</i> Flow Manager validates the time-window constraint (SHACL/SPARQL) using constraints from the Instantiated Constraint Cache.
After acceptance, the dishwasher enters the waiting state for $P2$ and keeps $P2$ selected.	<i>Observable evidence:</i> state change to $P2_WAITING$ (or equivalent state label); updated selected sequence in the announced plan/confirmation. <i>Framework component:</i> Flow Manager validates the post-acceptance state/plan conditions (SHACL/SPARQL).
If accepted, execution starts at $22:00 \pm 1$ minute.	<i>Observable evidence:</i> observed start timestamp (and the configured start time). <i>Framework component:</i> Time Manager provides temporal events/timestamps to the execution context; Flow Manager validates the start-time constraint (SHACL/SPARQL).
If accepted, the cycle completes by 23:30.	<i>Observable evidence:</i> observed end timestamp; run status indicates completion (e.g., COMPLETED). <i>Framework component:</i> Time Manager provides time context; Flow Manager validates the completion deadline and completion status (SHACL/SPARQL).

Building on the behavioral rules (test criteria) in Table 1, the proposed framework operationalizes behavioral interoperability testing by instantiating and executing a session-specific behavioral flow for the selected CAI. As shown in Figure 3, the Behavioral Test Manager (Steps 1-6) performs four functions for each session: (i) it aligns the CAI with the reference ontology to derive the relevant ontology fragment, (ii) it retrieves the corresponding behavioral template from the Task and Shape Library, (iii) it binds scenario parameters and identifiers using the ontology fragment and SUT runtime context (e.g., $P1$, $P2$, time windows, and state labels), and (iv) it stores the instantiated flow artefacts and associated SHACL/SPARQL constraint definitions in the Instantiated Constraint Cache, together with the session metadata needed for traceability. During execution (Steps 7-13), the Flow Manager retrieves the instantiated artefacts from the cache, maintains the evolving RDF state graph S_i , and evaluates the applicable constraint set C_i through pre-condition checks, transition execution (SPARQL-based operations), and post-condition checks. Time-bounded criteria are supported by the Time Manager, which provides tick and deadline events and injects timestamps into the execution context so that start-time and completion constraints can be evaluated against observed events. All validation outcomes, state identifiers, applied constraints, and timestamps are recorded in the RDF Trace Graph (Steps 14-16), which provides the trace evidence referenced in Table 1 and enables a traceable behavioral

verdict that complements semantic conformance checking. Overall, this example shows how ontology-grounded concepts such as the appliance, selectable sequences, observed state, request time, start/end timestamps, and completion status are bound through the CAI and behavioral template into executable SHACL/SPARQL checks and evaluated over the RDF state and trace graphs to produce each behavioral verdict.

5. Conclusion and Future Work

This paper introduced a conceptual framework for incorporating behavioral interoperability testing into the ontology-driven interoperability validation process previously proposed in [20]. By combining the assessment of semantic conformance and execution behavior, the proposed framework is intended to address the limitations of evaluating these two dimensions independently. Within an Energy Data Space context, this unified approach supports the validation of coordinated services, such as the flexible activation and optimization of different energy assets by an energy management system, not only with respect to correct semantic interpretation but also with respect to agreed operational constraints during interactions, including request-time windows, state transitions, and start/end time bounds. Embedding behavioral interoperability testing into the ontology-driven validation process provides a structured basis for evaluating both semantic and behavioral aspects of interoperability in practical Energy Data Space implementations. The proposed framework is illustrated through a simplified use case inspired by the JRC Code of Conduct for Energy Smart Appliances, whereas broader interaction scenarios are not addressed. From a practical perspective, the proposed integration may help reduce the effort required for manual functional testing by checking ordering and timing constraints directly from runtime traces (i.e., exchanged messages and observed system events with timestamps). This may support faster development and testing of Energy Smart Appliances in interoperability scenarios inspired by the JRC Code of Conduct (CoC). As more appliances are expected to comply with the CoC in the coming years, such an integrated testing approach could support future validation activities during development. A limitation of the approach is that the constraints and templates used for validation need to be updated when interoperability requirements evolve. Future work will focus on implementing and evaluating the workflow, including expressiveness and scalability. It will also extend the approach toward policy interoperability by expressing policy constraints as validation conditions over exchanged data and traces, thereby strengthening assurance across syntactic, semantic, behavioral, and policy layers.

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Declaration on Generative AI

During the writing of this paper, the author(s) used Grammarly and QuillBot for grammar and spelling checking, and minor rephrasing for clarity. After using these tools, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the publication's content.

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